

Cheat sheet - 2nd DSP exam/UFPA

Chapter 3 – Systems

Convolution in discrete (DT) and continuous (CT) time:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]. \quad (1)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau. \quad (2)$$

Convolution properties:

- Commutative: $a*b = b*a$
- Associative: $a*(b*c) = (a*b)*c$
- Distributive: $a*(b+c) = a*b + a*c$
- Duration: $N_1 + N_2 - 1$ samples
- Indices: if $a[n] = b[n]*c[n]$, the valid products to obtain $a[n_a]$ are $b[n_b]c[n_c]$ with $n_a = n_b + n_c$

$$y(t) = x_1(t)*x_2(t) = \mathcal{F}^{-1}\{X_1(f)X_2(f)\} \quad (3)$$

$$y[n] = x_1[n]*x_2[n] = \text{DTFT}^{-1}\{X_1(e^{j\Omega})X_2(e^{j\Omega})\} \quad (4)$$

Periodic or circular convolution (both signals of period T):

$$x_1(t) \otimes x_2(t) = \frac{1}{T} \int_{\langle T \rangle} x_1(\tau)x_2(t-\tau)d\tau \quad (5)$$

In discrete-time frequency domain (implicit period 2π):

$$x_1[n]x_2[n] \Leftrightarrow X(e^{j\Omega}) \otimes X_2(e^{j\Omega}) \quad (6)$$

$$y[n] = x_1[n] \otimes x_2[n] = \text{FFT}^{-1}\{X_1[k]X_2[k]\} \quad (7)$$

Complex exponentials are eigenfunctions of LTI systems. In continuous-time (also valid for DT):

$$e^{s_0 t} \rightarrow \boxed{\mathcal{H}} \rightarrow H(s_0)e^{s_0 t}.$$

Tabela 1: Nomenclature of special frequencies in filtering.

Symbol	Description
ω_c or f_c	Cutoff frequency: gain falls by $1/\sqrt{2}$ (−3 dB)
ω_n	Natural frequency, for example, of a resonator
ω_0	Center frequency of a pole
f_p	Passband frequency
f_r	Stopband (or rejection) frequency
F_s	Sampling frequency (in Hz or samples per sec.)
$F_s/2$	Nyquist (or folding) frequency

Quality factor of poles in CT: $Q \stackrel{\text{def}}{=} \frac{f_n}{\text{BW}}$ and DT: $Q = \frac{\omega_n}{2\alpha}$.

Second-order Butterworth filter:

$$H(s) = \frac{\omega_0^2}{s^2 + \sqrt{2}\omega_0 s + \omega_0^2}.$$

Group delay in CT (DT is similar):

$$\tau_g(\omega) = -\frac{d\Theta}{d\omega}. \quad (8)$$

If the deviation in passband should be within $[1 - D_p, 1]$ (linear scale) the ripple in dB to be within $[-R_p, 0]$ is:

$$R_p = -20 \log_{10}(1 - D_p).$$

For stopband:

$$R_s = -20 \log_{10} D_s.$$

Filter frequency scaling of a prototype $H(s)$ with cutoff frequency 1 rad/s:

$$G(s) = H(s)|_{s=s/\omega_c} \quad (9)$$

such that ω_c is the cutoff frequency of the new filter $G(s)$. Simplified notations for A/DT{·} conversion:

$$x[n] = x(t)|_{t=nT_s} \quad (10)$$

DT signal $x[n]$ with DTFT $X(e^{j\Omega})$ is converted into a sampled signal $x_s(t)$ with Fourier transform $X_s(\omega)$ via a DT/S conversion,

$$X_s(\omega) = X(e^{j\Omega})|_{\Omega=\omega T_s} = X(e^{j\omega T_s}). \quad (11)$$

$$H_s(\omega) = H_z(e^{j\Omega})|_{\Omega=\omega T_s} = H_z(e^{j\omega T_s}), \quad (12)$$

Perfect reconstruction of $x[n]$ when $h(t)$ is a sinc:

$$x(t) = \text{DT/S}\{x[n]\} * h(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t}{T_s} - n\right). \quad (13)$$

Power relation:

$$\mathcal{P}_s \stackrel{\text{def}}{=} \frac{\mathcal{P}_d}{T_s}. \quad (14)$$

IIR design methods based on $H(s)$: a) Matched Z-transform or pole-zero matching: $z_i = e^{s_i T_s}$. b) Impulse invariance

$$H(z) = T_s \mathcal{Z}\{h(t)|_{t=nT_s}\} \quad (15)$$

c) Step invariance:

$$\mathcal{Z}^{-1}\left\{H(z)\frac{1}{1-z^{-1}}\right\} = \mathcal{L}^{-1}\left\{H(s)\frac{1}{s}\right\}\Bigg|_{t=nT_s} \quad (16)$$

d) Backward difference: $s \leftarrow (1 - z^{-1})/T_s$

e) Forward difference: $s \leftarrow (1 - z^{-1})/(T_s z^{-1})$

f) Bilinear (or Tustin's): $s \leftarrow 2F_s \frac{z-1}{z+1}$

Bilinear frequency warping

$$\omega = \frac{2}{T_s} \tan\left(\frac{\Omega}{2}\right) = 2F_s \tan\left(\frac{\Omega}{2}\right), \quad (17)$$

$$\omega_a = \frac{2}{T_s} \tan\left(\frac{T_s \omega_d}{2}\right). \quad (18)$$

To match a frequency:

$$F'_s = \frac{\omega_m}{2 \tan\left(\frac{\omega_m}{2F_s}\right)}, \quad (19)$$

In FIR design via windowing, ideal lowpass filter impulse response:

$$h_{\text{LP}}[n] = \frac{\sin(\Omega_c n)}{\pi n} \quad (20)$$

Impulse response for the ideal highpass filter:

$$h_{\text{HP}}[n] = \delta[n] - h_{\text{LP}}[n]. \quad (21)$$

Symmetric FIR has linear phase and can be decomposed as

$$H(e^{j\Omega}) = e^{-j\Omega \tau_g} A(\Omega)$$

Realizations of digital filters: direct form I, direct form II, transposed direct form II, SOS in cascade, SOS in parallel.

Chapter 4 – Spectral Analysis

Windowing $x_w[n] = x[n]w[n]$ corresponds to *periodic convolution* in frequency-domain:

$$X_w(e^{j\Omega}) = X(e^{j\Omega}) \otimes W(e^{j\Omega}), \quad (22)$$

Hamming window:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \dots, N-1. \quad (23)$$

“Periodic” *Hann* window

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \dots, N-1. \quad (24)$$

Important performance metrics of windows: a) width of its main lobe, b) amplitude of its sidelobes, c) rate of sidelobe decay as frequency increases, d) and scalloping loss.

Expressions to investigate bin and non-bin-centered sinusoids, where $\alpha \in \mathbb{R}$ and $\Delta\Omega = 2\pi/N_{\text{fft}}$:

$$x[n] = A \cos\left(\alpha \frac{2\pi}{N_{\text{fft}}} n\right) \quad \text{and} \quad x[n] = A \cos(\alpha \Delta\Omega n). \quad (25)$$

DTFT $X_w(e^{j\Omega})$ of the windowed cosine $x_w[n] = x[n]w[n]$:

$$X_w(e^{j\Omega}) = \frac{A}{2} \left[\frac{\sin\left(\frac{N(\Omega+\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega+\Omega_c}{2}\right)} e^{-j\frac{(\Omega+\Omega_c)(N-1)}{2}} + \frac{\sin\left(\frac{N(\Omega-\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega-\Omega_c}{2}\right)} e^{-j\frac{(\Omega-\Omega_c)(N-1)}{2}} \right]. \quad (26)$$

Leakage is the redistribution of spectral energy caused by windowing a signal.

Picket-fence effect is the phenomenon whereby the DFT/FFT observes the continuous DTFT only at the FFT bin centers.

Scalloping loss is the reduction in the measured amplitude of a spectral component when its frequency lies between two adjacent DFT/FFT bin centers rather than at a bin center. Improving FFT resolution $\Delta f = F_s/N_{\text{fft}}$ by increasing N_{fft} cannot recover the leakage that occurred due to windowing.

Tabela 2: Energy spectral density (ESD) functions. The units of $\mathcal{G}(f)$, $\mathcal{G}(\omega)/(2\pi)$ and $\mathcal{G}(e^{j\Omega})/(2\pi)$ are J/Hz, J/(rad/s) and J/rad, respectively.

Time	ESD definition	Property
Cont.	$\mathcal{G}(f) = X(f) ^2$	$E = \int_{-\infty}^{\infty} \mathcal{G}(f) df$
Cont.	$\mathcal{G}(\omega) = X(\omega) ^2$	$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{G}(\omega) d\omega$
Disc.	$\mathcal{G}(e^{j\Omega}) = X(e^{j\Omega}) ^2$	$E = \frac{1}{2\pi} \int_{<2\pi>} \mathcal{G}(e^{j\Omega}) d\Omega$

Tabela 3: Power spectral density (PSD) functions. The units of $S(f)$, $S(\omega)/(2\pi)$ and $S(e^{j\Omega})/(2\pi)$ are W/Hz, W/(rad/s) and W/rad, respectively.

Time	Definition	Main property
Cont.	$S(f) = \mathcal{F}_f\{R(\tau)\}$	$\mathcal{P} = \int_{-\infty}^{\infty} S(f) df$
Cont.	$S(\omega) = \mathcal{F}_\omega\{R(\tau)\}$	$\mathcal{P} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) d\omega$
Discr.	$S(e^{j\Omega}) = \mathcal{F}\{R[\ell]\}$	$\mathcal{P} = \frac{1}{2\pi} \int_{<2\pi>} S(e^{j\Omega}) d\Omega$

PSD of continuous-time periodic deterministic signal:

$$S(f) = \sum_{k=-\infty}^{\infty} |c_k|^2 \delta(f - kF_0), \quad (27)$$

with Fourier Series coefficients c_k and period $T_0 = 1/F_0$. PSD of discrete-time sinusoid $x[n] = A \cos(\Omega_1 n)$:

$$S(e^{j\Omega}) = \sum_{p=-\infty}^{\infty} \frac{\pi A^2}{2} [\delta(\Omega + \Omega_1 + p2\pi) + \delta(\Omega - \Omega_1 + p2\pi)]$$

For a WSS random process $\mathcal{X}(t)$ with PSD $S_x(f)$, the PSD of $\mathcal{Y}(t) = \mathcal{X}(t)e^{j2\pi f_c t}$ is $S_y(f) = S_x(f - f_c)$.

Mean-square spectrum (MS spectrum):

$$S_{\text{ms}}[k] = |\text{DTFS}\{x[n]\}|^2. \quad (28)$$

If a WSS process $\mathcal{X}(t)$ is the input of a LTI system $\mathcal{X}(t) \rightarrow \boxed{h(t)} \rightarrow \mathcal{Y}(t)$, then $S_y(f) = |H(f)|^2 S_x(f)$. In discrete-time: $S_y(e^{j\Omega}) = |H(e^{j\Omega})|^2 S_x(e^{j\Omega})$.

The *bilateral* PSD of a white noise is

$$S(f) = \mathcal{F}\left\{\frac{\mathcal{N}_0}{2} \delta(\tau)\right\} = \frac{\mathcal{N}_0}{2}, \quad \forall f \quad (29)$$

and its average power over a bandwidth BW is $\mathcal{P} = \mathcal{N}_0 \text{BW}$.

For discrete-time white noise: $S_y(e^{j\Omega}) = \sigma_x^2 |H(e^{j\Omega})|^2$.

Periodogram is an approximation of $S(f)$ given by

$$\hat{S}[k] \stackrel{\text{def}}{=} \frac{1}{N \text{BW}} |\text{FFT}\{x_N[n]\}|^2. \quad (30)$$

$$\mathcal{P} \approx \Delta f \sum_{k=0}^{N-1} \hat{S}[k], \quad (31)$$

Relation between MS spectrum and PSD:

$$\hat{S}_{\text{ms}}[k] = \mathcal{P}_k = \Delta f \hat{S}[k]. \quad (32)$$

$$\hat{S}(e^{j\Omega}) = \sum_{k=-N+1}^{N-1} \hat{R}[k] e^{-j\Omega n}. \quad (33)$$

Learn to use Welch’s method in Matlab:

```
[Sx,w]=pwelch(x>window,Num_overlap,N_fft,Fs,
'twosided')
```

and Python

```
from scipy.signal import welch
w,Sx=welch(x>window,nperseg=len(window),
noverlap=Num_overlap,nfft=N_fft,
window=window,return_onesided=False)
```

Auto-regressive modeling: find prediction-error FIR filter $A(z) = 1 + \sum_{i=1}^P a_i z^{-i} = 1 - \hat{A}(z)$, where $\hat{A}(z) = -\sum_{i=1}^P a_i z^{-i}$ provides the predictions

$$\tilde{y}[n] = -\sum_{i=1}^P a_i y[n-i], \quad (34)$$

such that $A(z)$ minimizes the power of the prediction error is $x[n] = y[n] - \tilde{y}[n]$. Having $A(z)$, the estimated PSD is:

$$\hat{S}(e^{j\Omega}) = |H(e^{j\Omega})|^2 = \frac{g^2}{|A(e^{j\Omega})|^2}, \quad (35)$$

Short-time Fourier transform (STFT):

$$X(\tau, f) = \int_{-\infty}^{\infty} x(t) w(t - \tau) e^{-j2\pi f t} dt \quad (36)$$